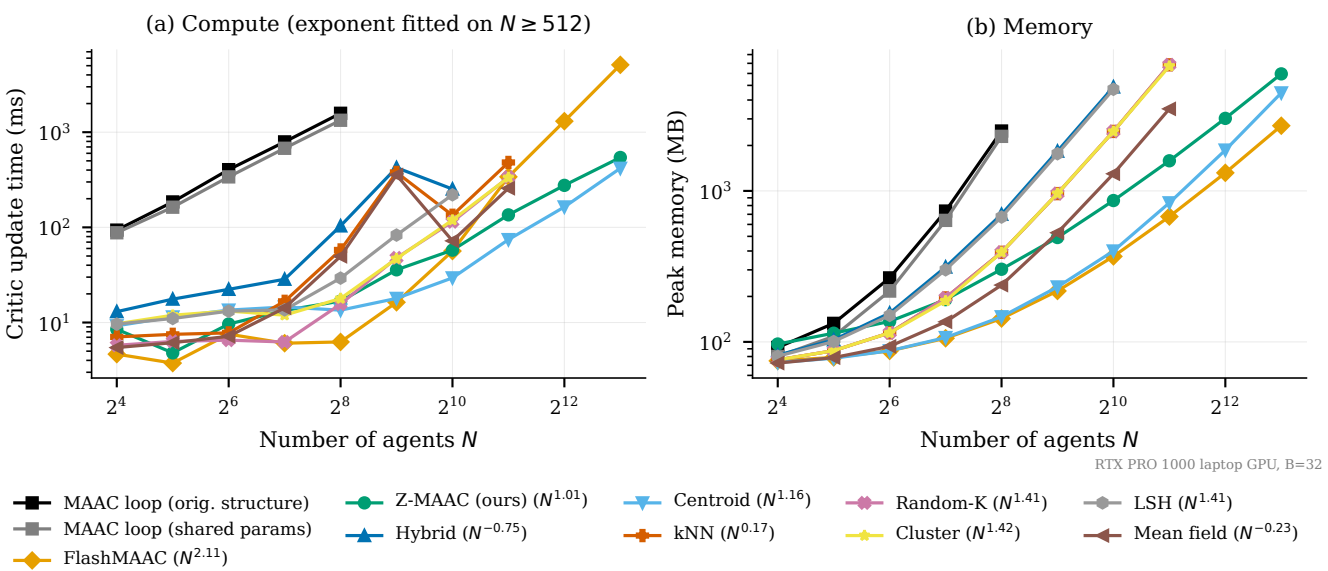
(a) Compute (exponent fitted on $N \geq 512$)
Critic update time (ms)
10^3
10^2
10^1
2^4
2^6
2^8
2^{10}
2^{12}
Number of agents N
(b) Memory
Peak memory (MB)
10^3
10^2
2^4
2^6
2^8
2^{10}
2^{12}
Number of agents N
RTX PRO 1000 laptop GPU, B=32
MAAC loop (orig. structure)
MAAC loop (shared params)
FlashMAAC ($N^{2.11}$)
Z-MAAC (ours) ($N^{1.01}$)
Hybrid ($N^{-0.75}$)
Centroid ($N^{1.16}$)
kNN ($N^{0.17}$)
Random-K ($N^{1.41}$)
Cluster ($N^{1.42}$)
LSH ($N^{1.41}$)
Mean field ($N^{-0.23}$)